

AI/ML Intern Interview with Vamshi (Sumedha Tatti)

Meeting started: Aug 29, 2026, 12:25:40 PM

Meeting duration: 7 minutes

Meeting participants: Sumedha Tatti, Vamshi Bharadwaj M V

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Detailed summary

Overview

The interview on 2026-08-29 was an AI/ML intern technical discussion between Vamshi Bharadwaj M V (interviewer) and Sumedha Tatti (candidate) focused on Sumedha's machine-learning project experience and a follow-up task: modeling IMU noise using non-Gaussian ML methods with a public dataset, to be submitted by Monday.

Key discussion points

- Candidate's primary project (neonatal cry analysis)
 - Problem: Continuous monitoring of neonates is difficult; detecting neonatal asphyxia early is critical to avoid long-term neurological damage.
 - Data: Used the Baby Chillanto (public) cry dataset containing ~847 cries: ~500 normal and ~347 asphyxia (numbers reported as "847 cries: 7? normal and 347 asphyxia" — candidate clarified use of ~847 total with a class split around 347 asphyxia).
 - Three signal representations evaluated:
 1. Handcrafted features — features based on perceptual audio scales

(Greenwood, mel-frequency cepstral-like coefficients and other biologically motivated transforms).

2. Pretrained audio embeddings — used pre-trained audio models (candidate named “vikish” and “gamnet” / likely VGGish and YamNet) and then downstream classification.

3. Spectrogram images — time-frequency spectrogram inputs to a classifier (ResNet-50 used).

- Models and pipelines:

- Pipeline A (handcrafted): classical ML models — SVM, Random Forest (RF), KNN, and a neural net (PNN/other small NN mentioned).

- Pipeline B (pretrained embeddings): embeddings from VGGish/YamNet; SVM used downstream due to small dataset size.

- Pipeline C (spectrograms): ResNet-50 for classification.

- Candidate’s capabilities:

- Comfortable with signal processing, feature extraction, and end-to-end ML for audio.

- Able to learn new models (e.g., wave transformers) and adapt across modalities because of electronics and biomedical background.

- Interviewer task assignment (IMU noise modeling)

- Task request: Develop an ML-based noise model for IMU (inertial measurement unit) sensors (accelerometer / gyroscope). Do not assume Gaussian noise; use ML approaches to model noise parameters.

- Requirements:

- Use an open-source/public dataset for IMU sensor data.

- Apply machine-learning techniques (not classical Gaussian noise assumptions) to model sensor noise characteristics.

- Analyze the data (e.g., using transforms such as wavelet or Fourier) as appropriate; interviewer suggested typical transforms are common but expects ML-based noise modeling beyond Gaussian assumptions.
- Deadline / follow-up:
 - Sumedha can revert by Monday with implementation and results.
 - Successful submission may lead to a next-round interview.

Agreements, action items, and next steps

- Action for Sumedha Tatti:
 - Select a suitable public IMU dataset (accelerometer and/or gyroscope).
 - Implement ML-based noise modeling approaches that do not assume Gaussian noise (possible directions: deep generative models, variational autoencoders, normalizing flows, mixture density networks, time-series models like LSTMs/transformers combined with probabilistic outputs, state-space learning, or nonparametric methods).
 - Include appropriate preprocessing and transforms (e.g., wavelet, Fourier, spectrograms) as part of the analysis pipeline, and justify choices.
 - Provide the implementation, results, and a brief report by Monday (as agreed).
- Action for Vamshi Bharadwaj M V:
 - Review the submitted noise-modeling work; if satisfactory, invite Sumedha to the next interview round.

Clarifications and candidate notes

- Candidate asked for time to complete the task and was granted until

Monday.

- Candidate expressed readiness to handle wave-transformer and other advanced signal-processing/ML techniques.
- Candidate's background: electronics engineering with an interdisciplinary pivot toward ML for healthcare applications.

Transcript

00:00 Vamshi B.M.V.: The good afternoon, I hope you can hear me.

00:06 Sumedha T.: Yes, I can hear it.

00:10 Vamshi B.M.V.: great so Mehta just

00:16 Sumedha T.: yeah

00:17 Vamshi B.M.V.: I'm sorry, I think I just click some leader da, but yeah, I hope you are doing great having a great time today.

00:26 Sumedha T.: Yes, yes, yeah.

00:29 Vamshi B.M.V.: Just give me a small you know kind of breathe nothing about your name and courage because I already have everything about you.

00:36 Sumedha T.: yeah

00:37 Vamshi B.M.V.: So tell me one thing about one of the best projects that you're worked on something that you would like to share.

00:44 Sumedha T.: So one of the projects best products so far in the ML domain has been in front

00:49 Sumedha T.: Hi, I'm transcribing this call with my Tactiq AI Extension.

<https://tactiq.io/r/transcribing> Hi, I'm transcribing this call with my Tactiq AI Extension. <https://tactiq.io/r/transcribing>

00:50 Sumedha T.: cry and analysis so the problem we were faced we came across in Healthcare domain was that for babies new needs. You can't continuously monitor 24 bar 7 by a human or a nurse because there are so many babies and you miss important physiological or important cry characteristics because you know you can't

01:17 Sumedha T.: monitor them always and one of the main problems with neonates was asphyxia detection which is something when is something

when a baby doesn't get enough oxygen to the brain and it can cause neurology. It's neurological problems later, so we thought of using cry as a biomarker and then analyzing the cry to detect whether the crime was from maybe or a normal baby.

01:49 Vamshi B.M.V.: Okay, how do you analyze that?

01:49 Sumedha T.: this is So we actually tried three different cried representations. We used baby cilantro database first because we had to train the model and we could not ask for clinical validation before showing anything so David database is a publicly available data. Set it has some 847 cries find out 7 normal and 347. You know as fixated Crays we took three representations first one was handcrafted features. So there are features like Greenwood substrual coefficients gametone frequency substrual coefficients mal frequency etc coefficients which are basically they scale the cry according to the human audio perception converts it into logarithmic scale and makes it more biologically compatible rather than just taking random features, so three major representations was for pipeline a we took these.

02:52 Sumedha T.: After it features and then pipeline B we used pre-trained audio embeddings like

02:53 Vamshi B.M.V.: okay

02:58 Sumedha T.: vikish and gamnet and saw whether the generalized sounds can fit to confirm to our signals better than the handcrafted features and then the third approach was using spectrograms where we convert the price into frequency versus time representations and then. Giving it to a classifier so for first approach for hand rest handcrafted features. We used classical ml models such as svm RFM and k n n p n n and then

03:32 Vamshi B.M.V.: okay

03:34 Sumedha T.: for pipeline we use vigish and amnet for embeddings and then classified them using spm because our data set was very small so down down the stream. We have to confirm it or data set so we use svm. And then for pipeline c we used resonate 50 for the classification classification.

04:00 Vamshi B.M.V.: Okay now you tell me one thing yeah okay. I understand so now you're doing

04:01 Sumedha T.: And the actual yeah.

04:05 Vamshi B.M.V.: teacher mapping right so with respect to the features.

You're trying to get the

04:08 Sumedha T.: Yes, yes.

04:09 Vamshi B.M.V.: details if I ask you to do a wave transformer things like that would be able to do it. On any kind of data.

04:18 Sumedha T.: I Yes, I can that's because I had to learn from scratch the signal processing and everything so I can do that.

04:31 Vamshi B.M.V.: You come from electronics domain. Why do you need?

04:36 Sumedha T.: Because I initially chose electronics in the medical domain because I was interested in that domain, but eventually when I learned everything when I got it problems domains and everything I realized that everything has to be interdisciplinary account for a particular problem you cannot you cannot focus on just single thing rather than it's a mix of everything every domain needs to come and then I eventually become interested in this ml domain for healthcare application. So that's how I improve my projects according to that when I became interested in that.

05:13 Vamshi B.M.V.: Yeah, then, let's do that one thing. I'll give you a problem statement. You

05:16 Sumedha T.: yeah

05:17 Vamshi B.M.V.: might have my email ID already because I am connected to agreement right so

05:17 Sumedha T.: okay yeah Yes. Yes.

05:24 Vamshi B.M.V.: essentially what I want is noise modelling related to imu.

05:28 Sumedha T.: okay

05:29 Vamshi B.M.V.: Can you write it down?

05:32 Sumedha T.: Yeah, I am writing it.

05:34 Vamshi B.M.V.: yeah How can you do it using ml that is your machine learning aspect I am you is electronics inertial measurement unit you might already know.

05:43 Sumedha T.: Yes. okay

05:48 Vamshi B.M.V.: And if you can implement using any open source dataset, how are you modeling? The noise parameters inside imu? I would like to look.

05:57 Sumedha T.: Do you need it now or can I have some time?

06:00 Vamshi B.M.V.: Yeah, I mean you can revert back on Monday it's not a problem.

06:04 Sumedha T.: okay, so you want me to take a publicly available data set and then do a noise

06:08 Vamshi B.M.V.: Yes.

06:10 Sumedha T.: modeling on the

06:11 Vamshi B.M.V.: Usually any transform you do a railway transform you analyze the data, but

06:17 Sumedha T.: okay

06:18 Vamshi B.M.V.: usually they use gaussian, but I don't want gaussian. I want some other machine learning based models for noise modelling the accelerometer or garethoped it.

06:27 Sumedha T.: Okay, Axel rometer of come again, sorry exotherm okay.

06:29 Vamshi B.M.V.: okay guys Yes okay madam, can I get it?

06:38 Sumedha T.: Yeah, okay, yes.

06:40 Vamshi B.M.V.: I think I think it was good conversing with you. So just come back with this.

06:43 Sumedha T.: yeah

06:44 Vamshi B.M.V.: We'll have a conversation once you come back if your submission is right, you will be called for the next single okay.

06:52 Sumedha T.: Okay, thank you. Is there anything you would suggest me to improve from this

06:54 Vamshi B.M.V.: thanks, thanks, Melody

06:58 Sumedha T.: interview?

07:00 Vamshi B.M.V.: Nothing nothing. I think we just started talking so I don't see anything at least on it. Keep going okay. Yeah.

07:05 Sumedha T.: okay thank you, so I'll be I'll

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